Containers & OpenShift Workshop

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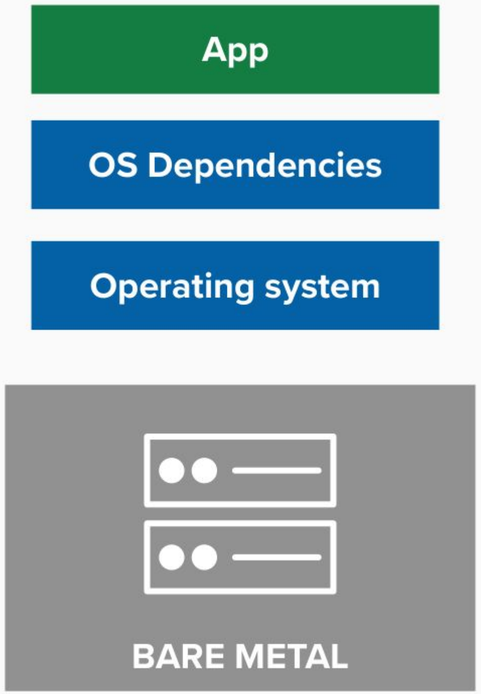
containers are created from container images

A LITTLE BIT ABOUT VIRTUALIZATION

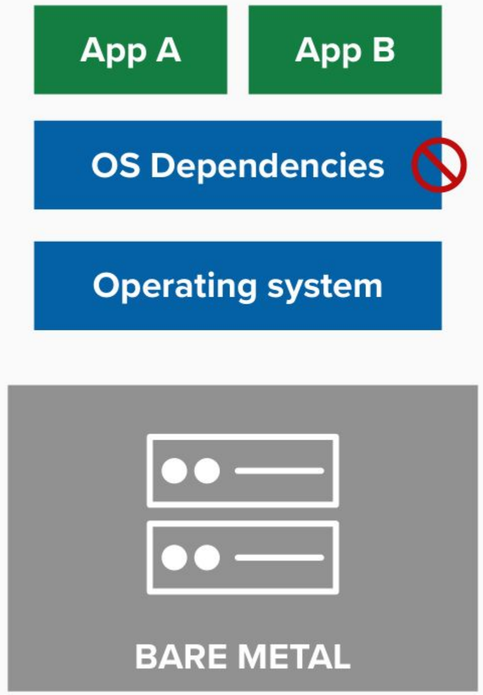


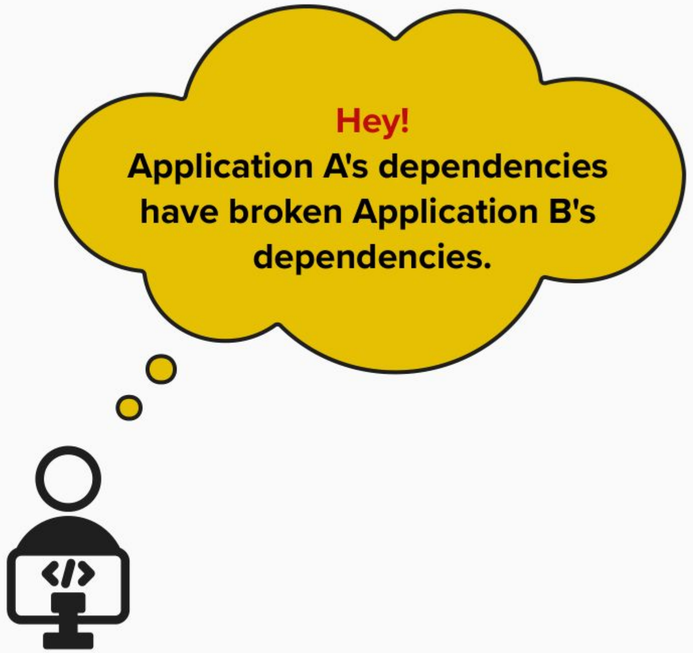


1. Traditional deployment



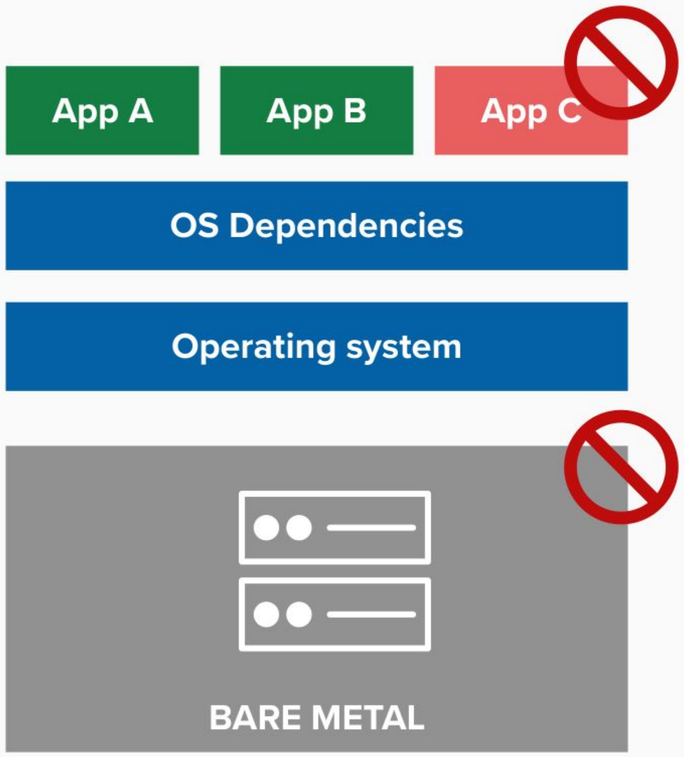
1. Dependency Management Challenge





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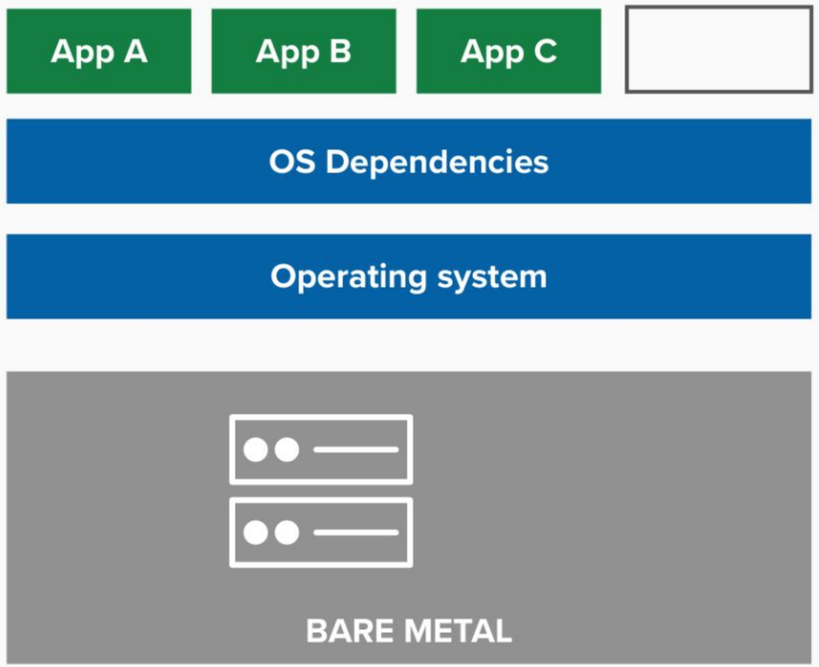
1. Scalability Challenge

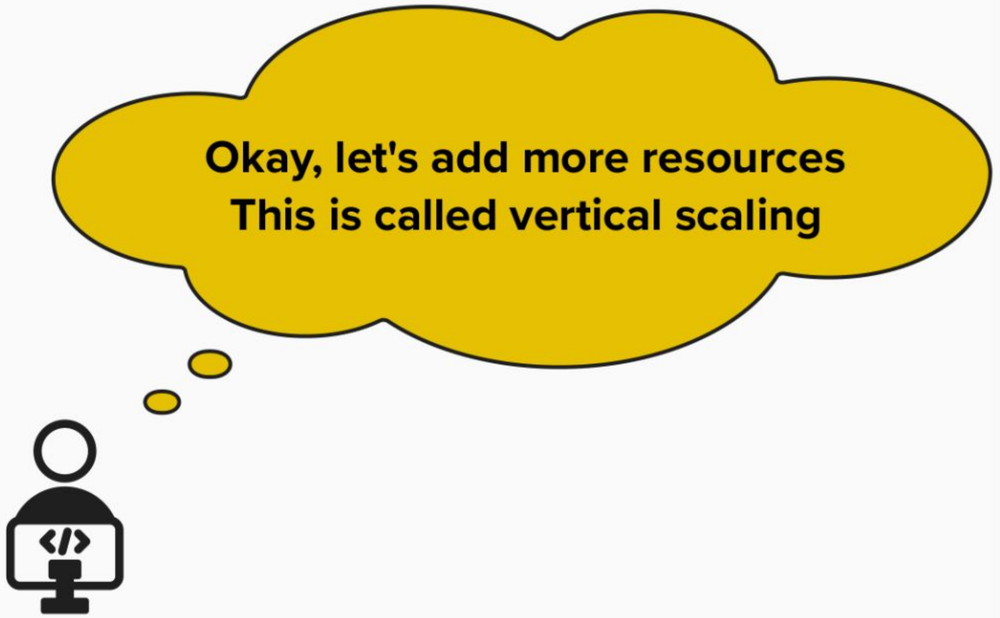




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1. Vertical scaling



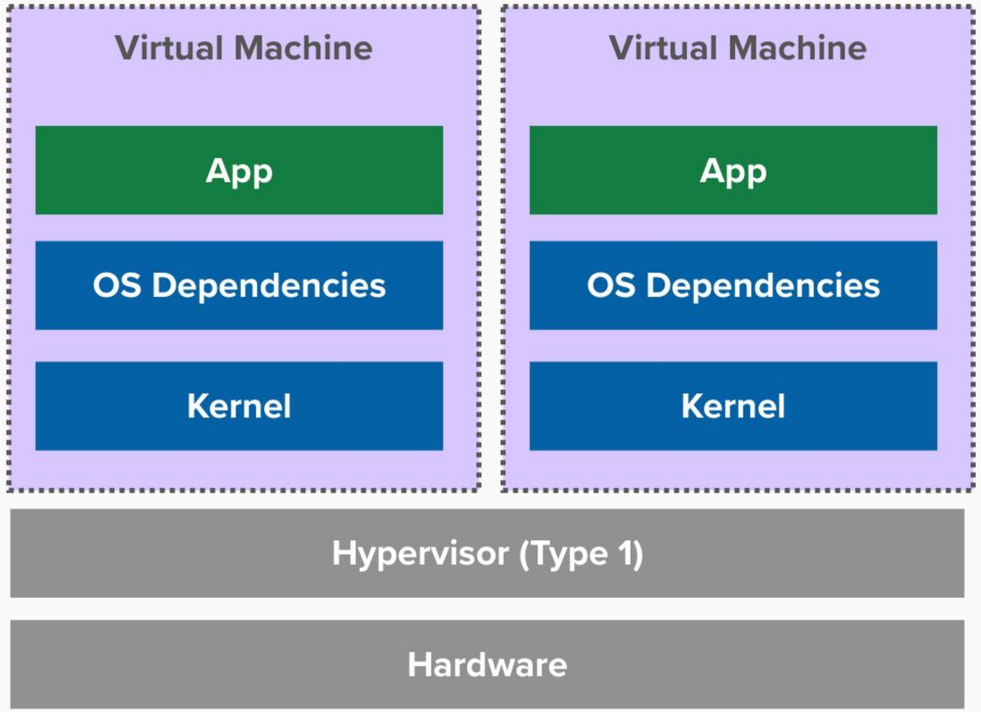


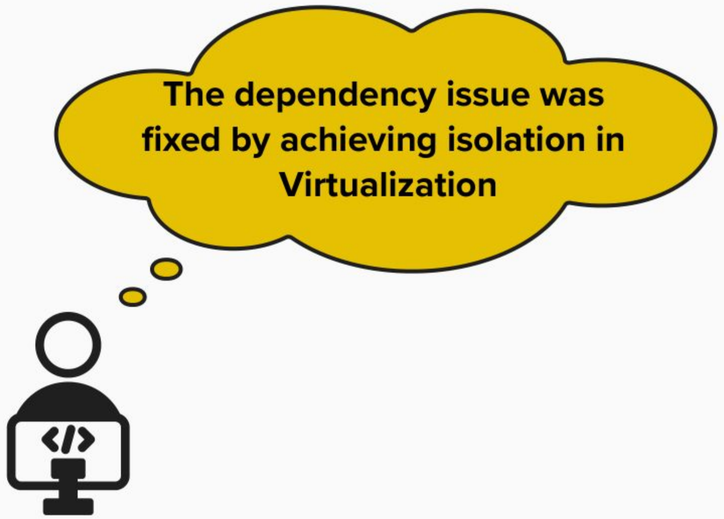
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1. Vertical scaling problem

Once we run out of resources again, we will need to keep adding more and more resources to the host. This approach is neither scalable nor maintainable.

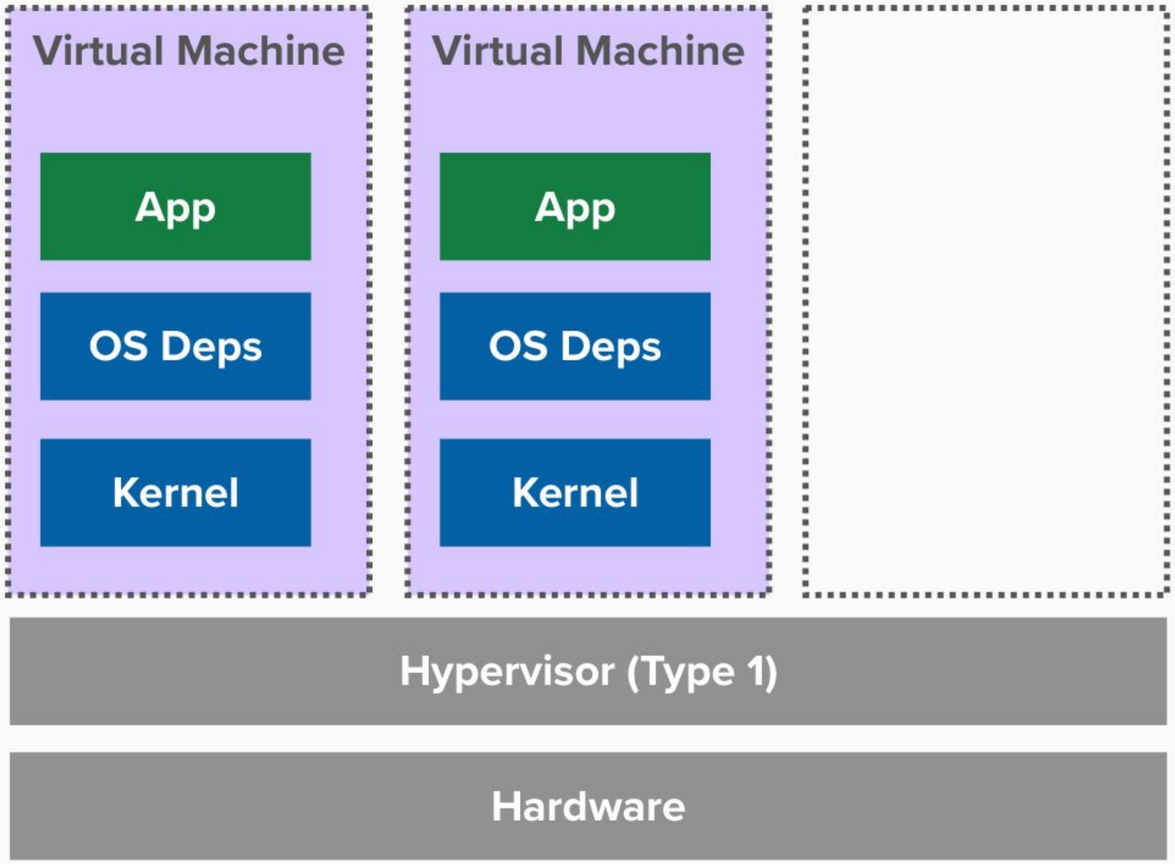
1. Virtualization as a Solution for Dependency Management Issues

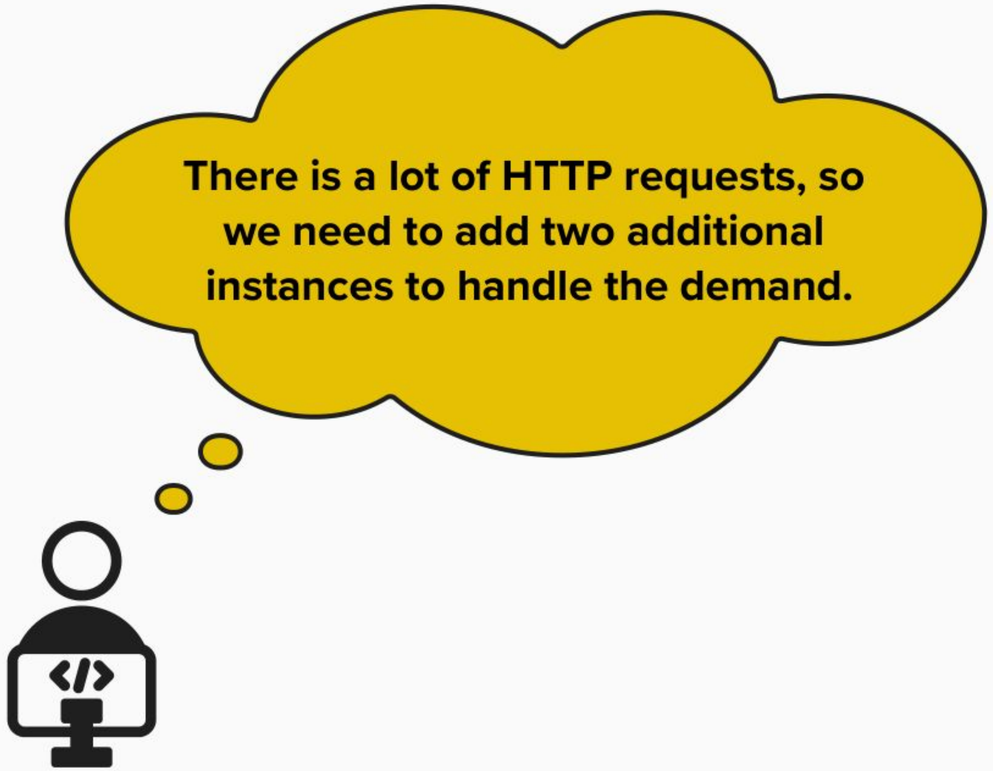




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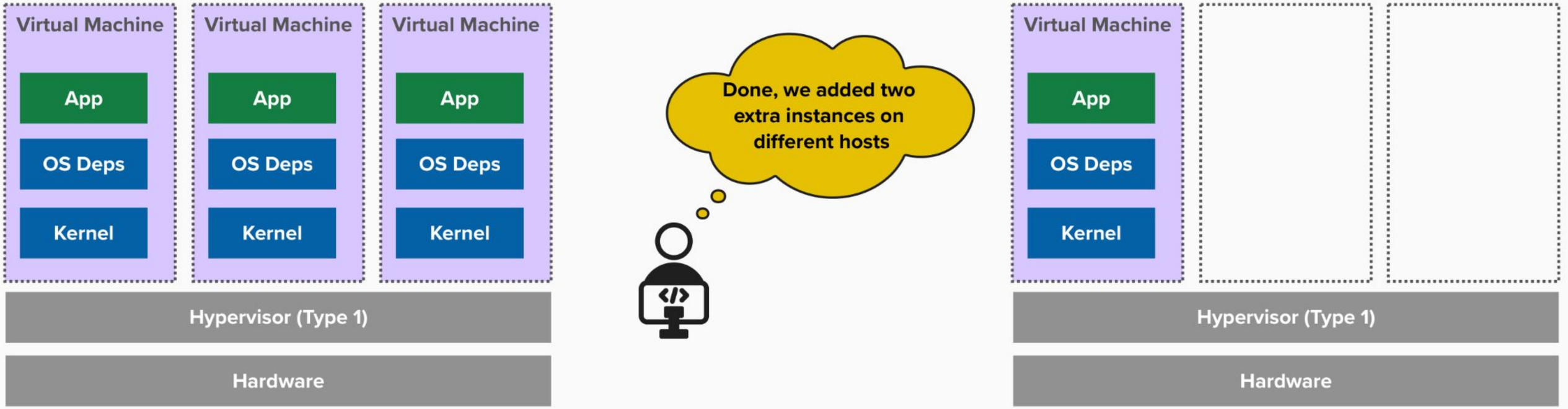
1. Virtualization as a Solution for scalability



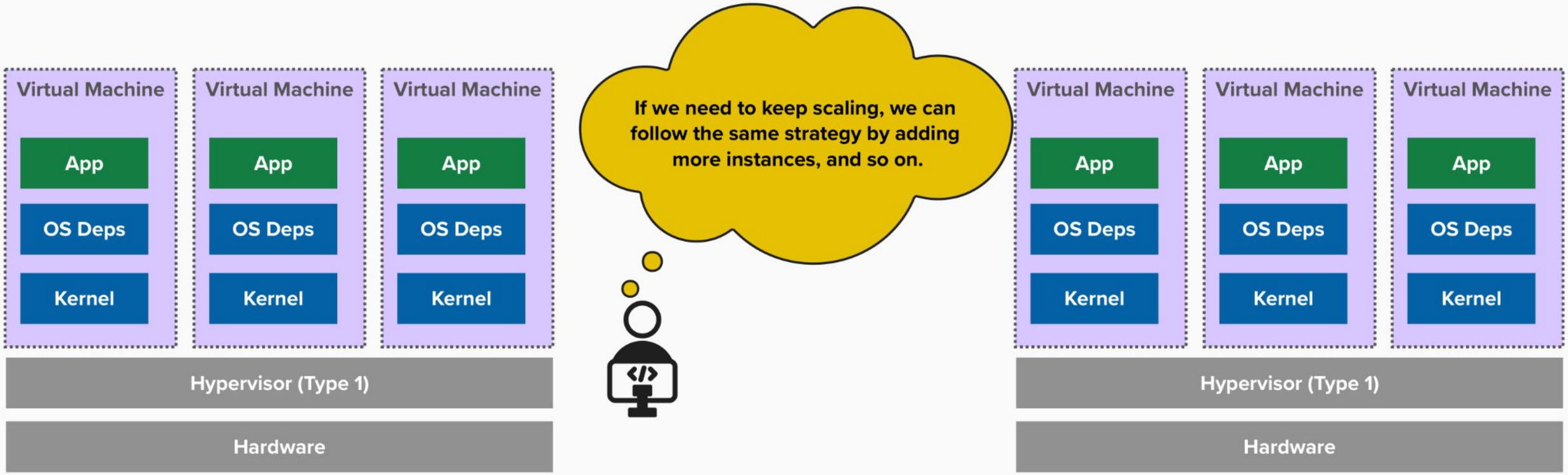


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1. Virtualization as a solution for scalability

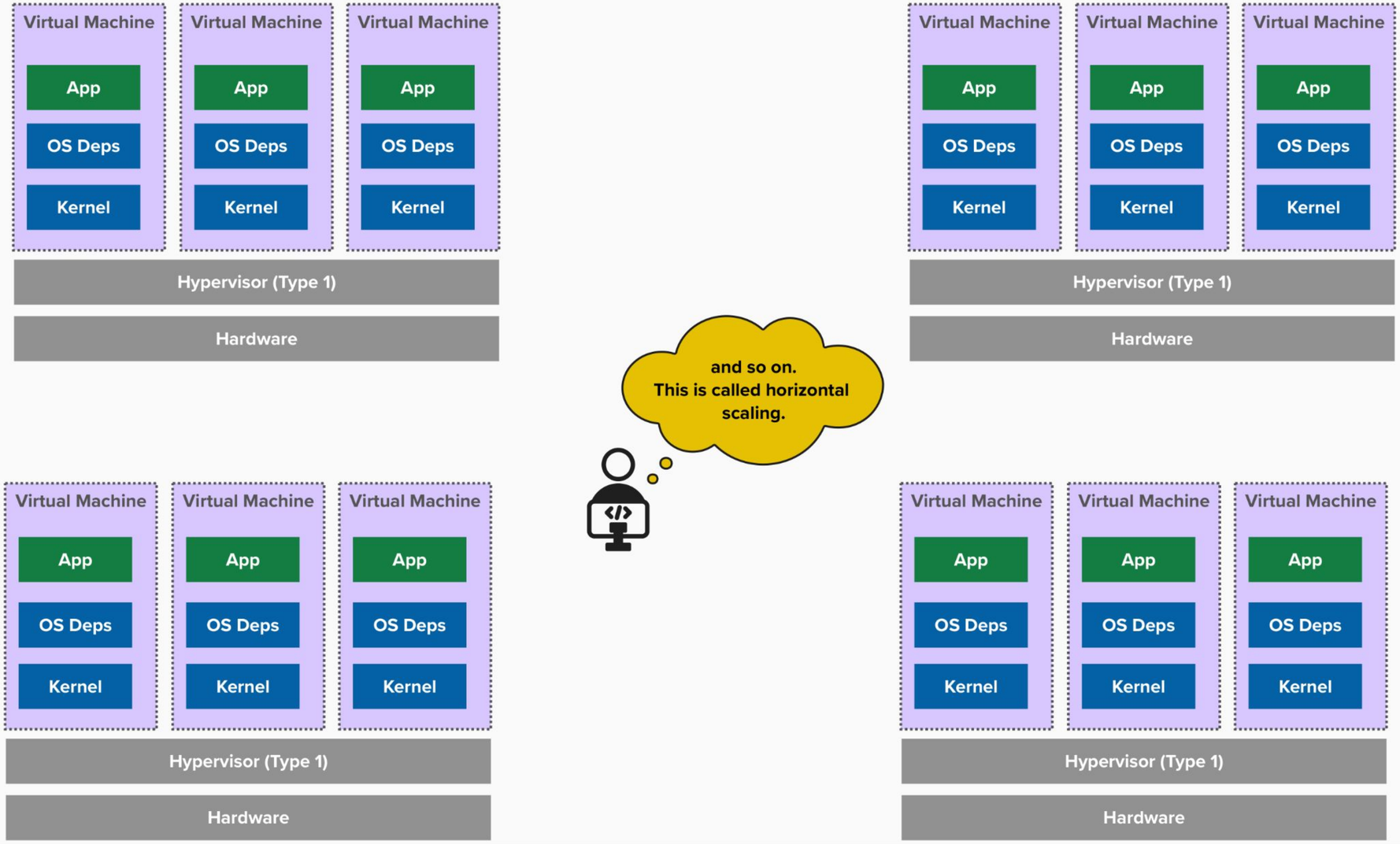


1. Virtualization as a solution for scalability



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1. Virtualization as a solution for scalability



1. Virtualization as a Solution for scalability

However, there are some challenges:

- How can we deploy additional application instances more quickly and reduce VM startup time?
- How can we ensure application portability when VMs are migrated between different hosts?
- How can we package application dependencies effectively to make VMs more lightweight?

containers are created from container images

Let’s do a benchmark! ⏱️

Containers vs Virtual Machines

containers are created from container images

CONTAINERIZATION AS A SOLUTION

2. What are containers?

- a **Set of Processes** sharing a common Kernel's resources for execution.
- **Isolated** from the rest of the machine.
  (can't see/affect/harm host or other containers)
- **Implementation:**
  - Uses **namespaces** to view the system's privately
    (network interfaces, PID tree, etc.)
  - Uses **cgroups** to have metered/limited/reserved resources.

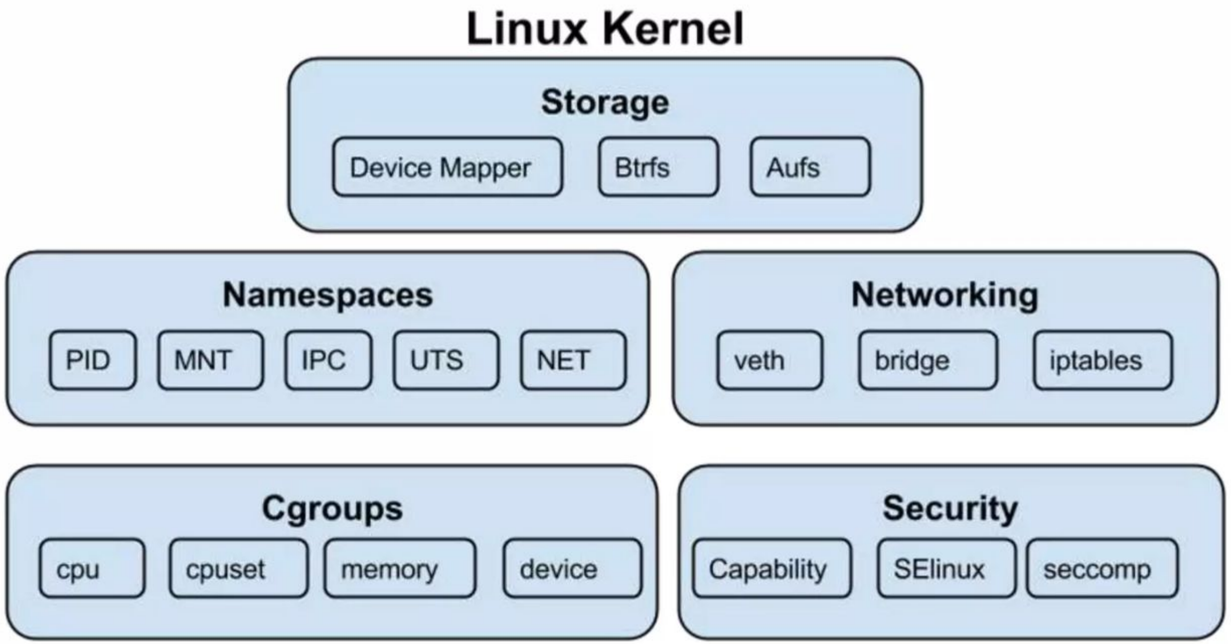
2. What are containers?

- **Fast**
  - Boots in milliseconds
- **Lightweight**
  - Just a few MBs of intrinsic disk/memory usage
- **Portable**
  - Migration from one system to another like a VM.
  - Deploy new applications or kill the old ones instantly.
- **Secure**
  - Considerable level of security to applications, as Containers remain isolated from each other on the guest OS.





Under the hood of Linux Containers



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2. What are containers?

It depends who you ask

Infrastructure

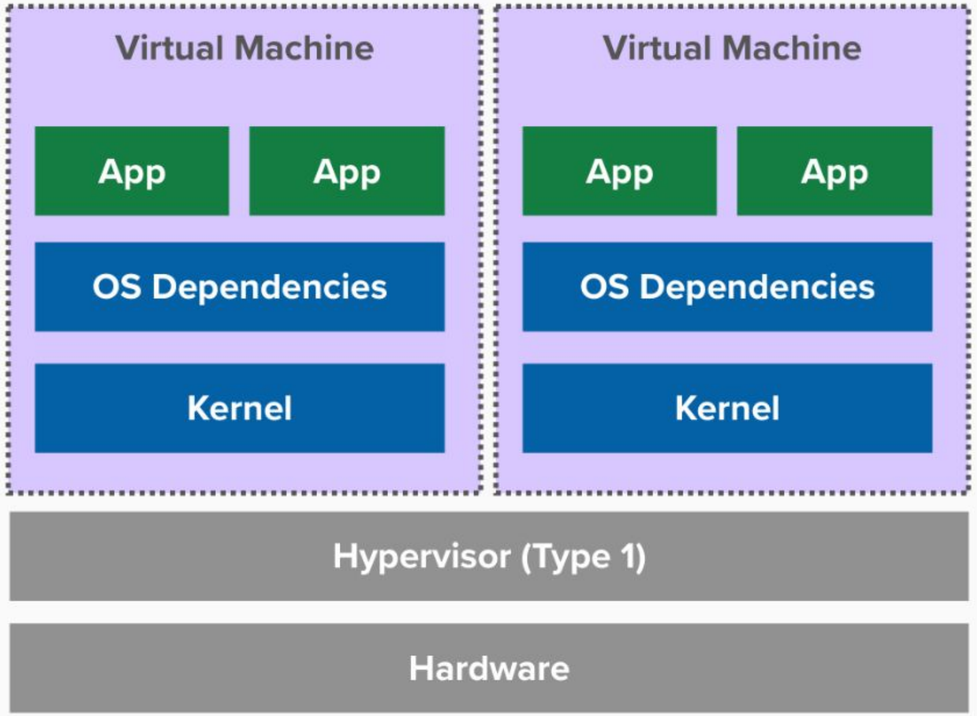
- Applications process on a shared kernel
- Simpler, lighter, and denser than VMS
- Portable across different environments

Application

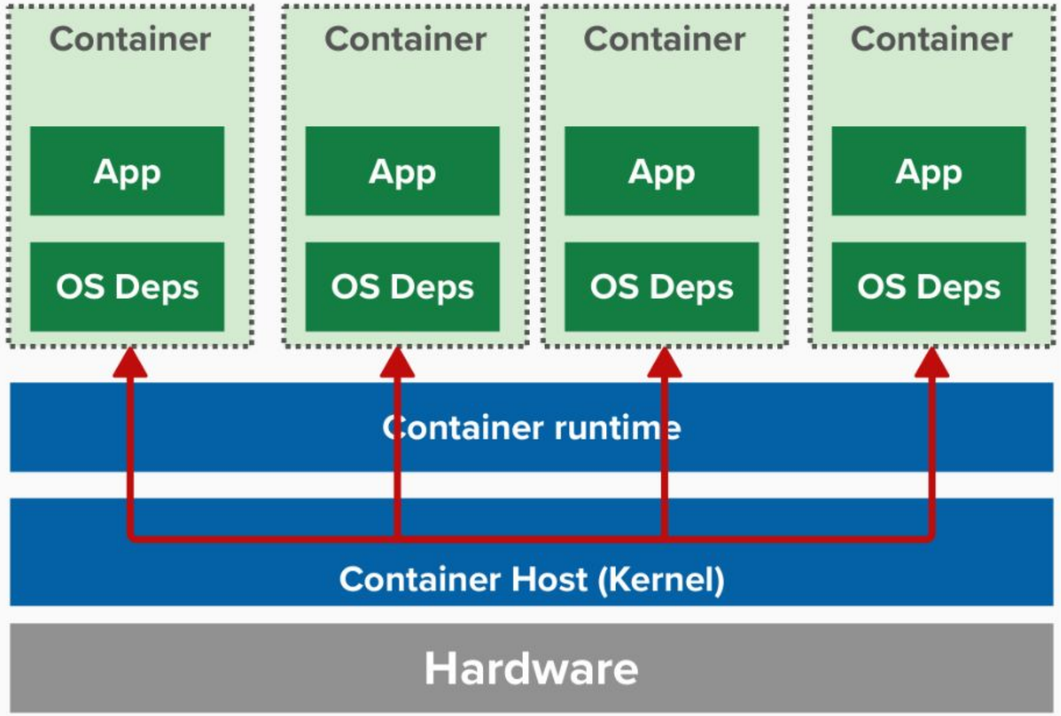
- Package apps with all dependencies
- Deploy to any environments in seconds
- Easily accessed and shared

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2. Faster than a virtual Machine



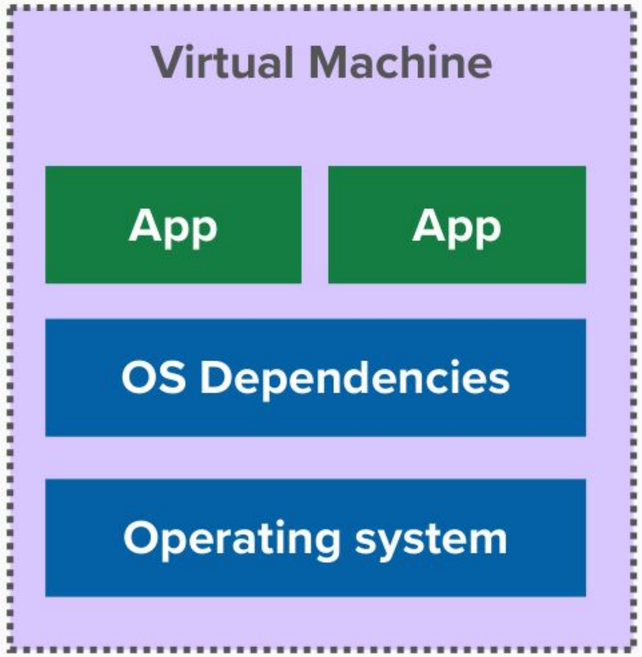
- Virtual machines are isolated
- App are not

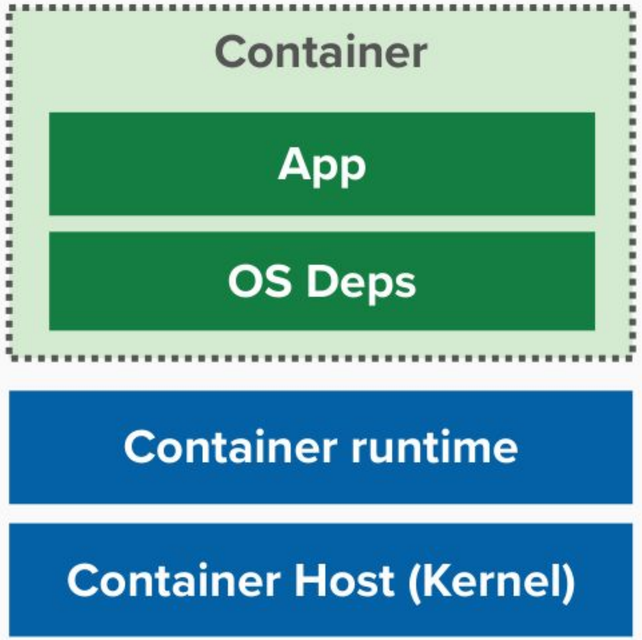


- Containers are isolated
- App are as well

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2. Virtual Machines and Containers



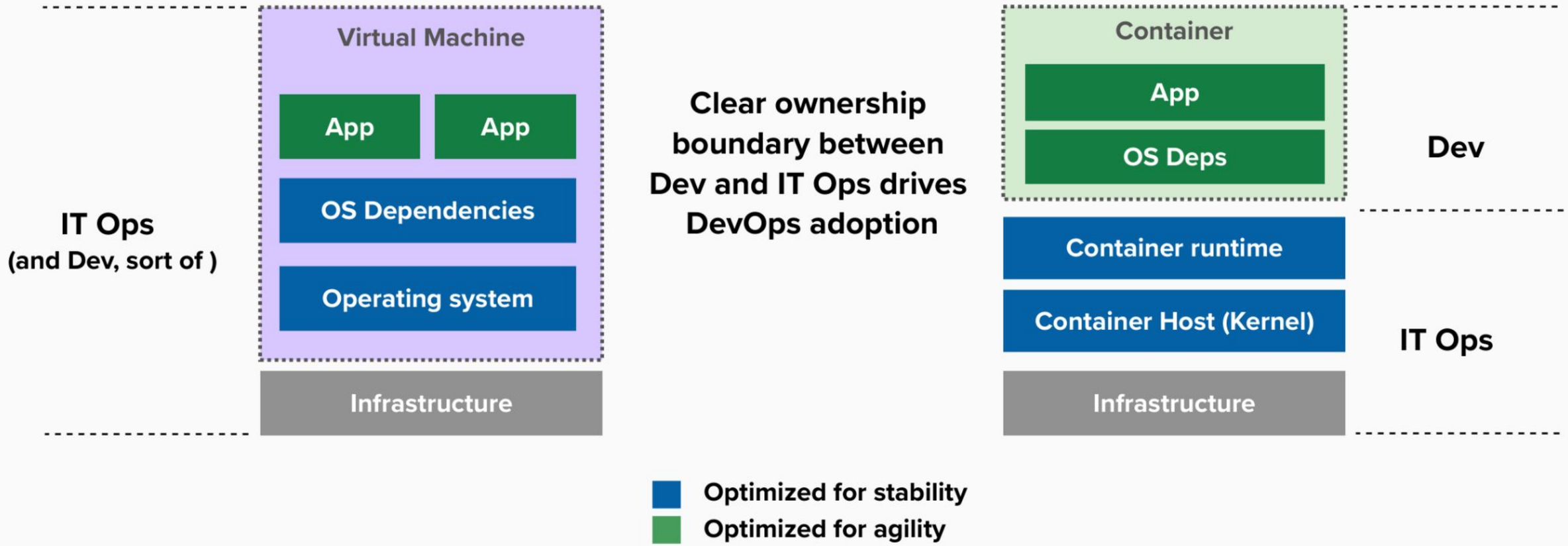


- ➕ VM Isolation
- ➖ Complete OS
- ➖ Static Compute
- ➖ Static Memory
- ➖ High Resource Usage

- ➕ Container Isolation
- ➕ Shared kernel
- ➕ Burstable Compute
- ➕ Burstable Memory
- ➕ Low Resource Usage

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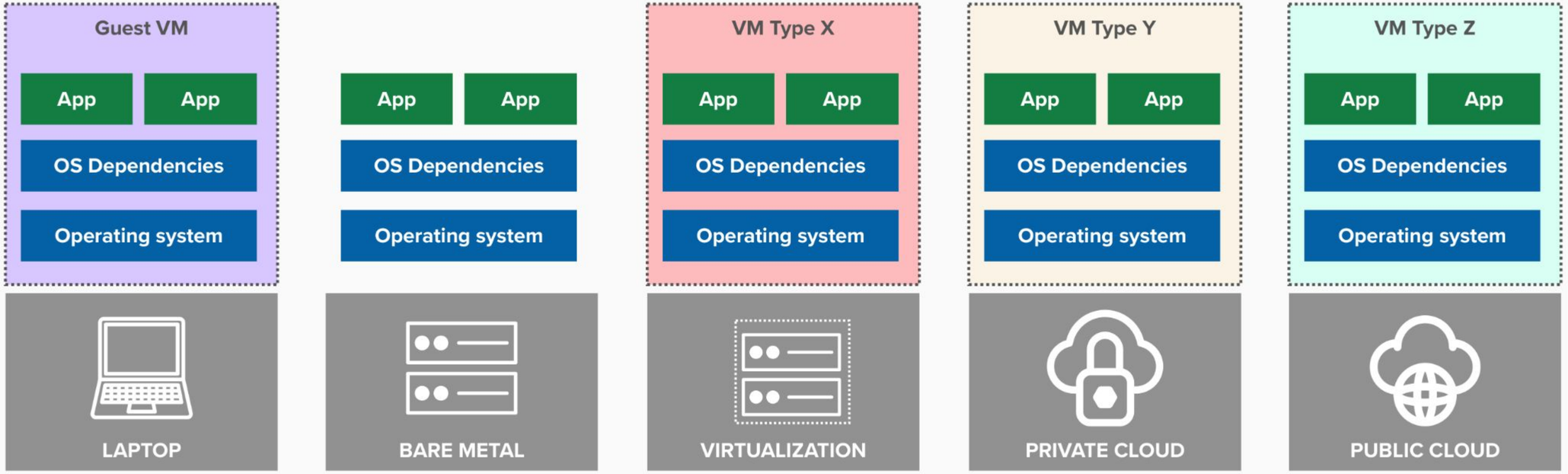
2. Virtual Machines and Containers



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2. Application Portability with VM

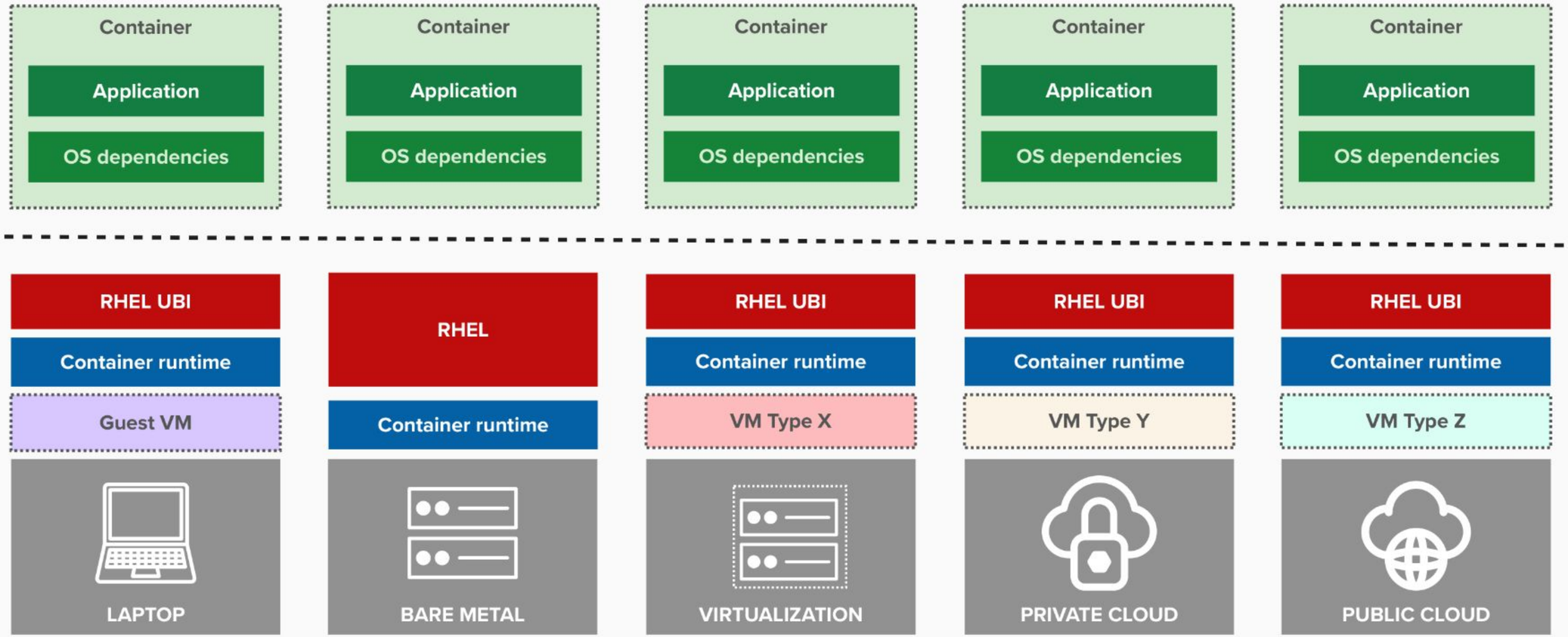
Virtual Machines are **NOT** portable across hypervisors and do **NOT** provide portable packaging for applications



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2. Application Portability with Containers

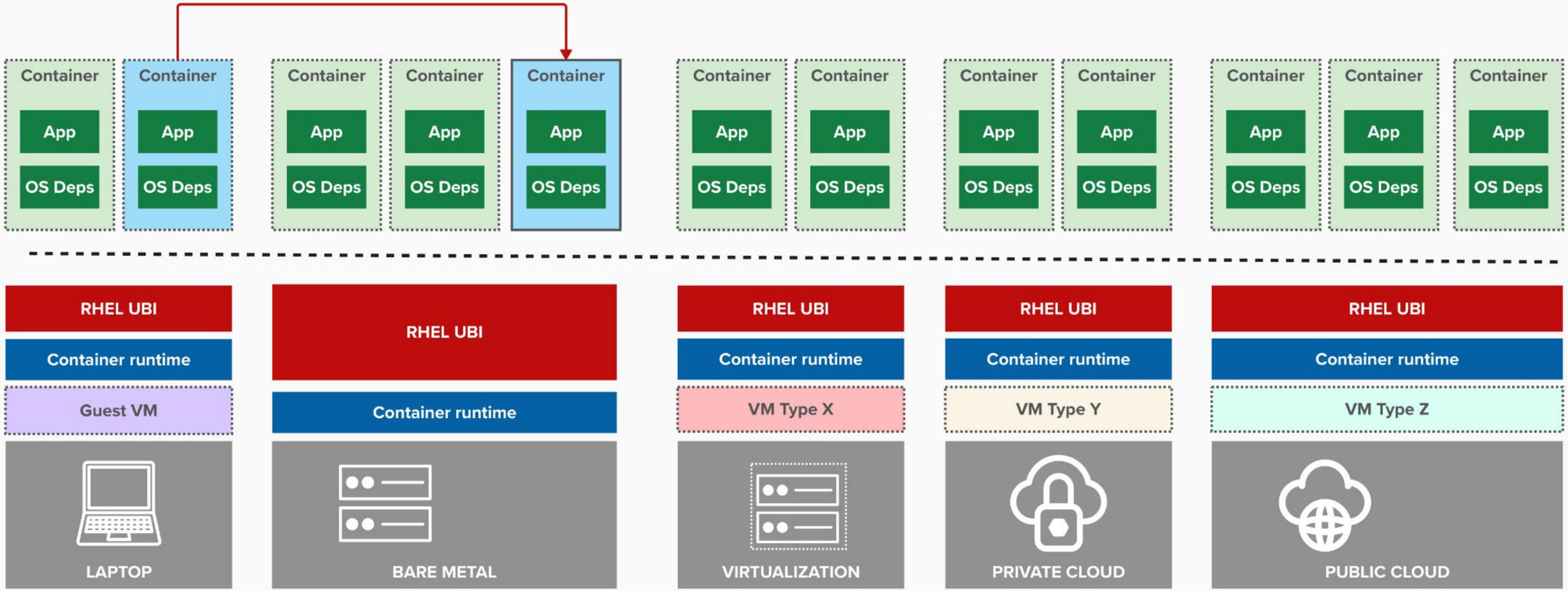
RHEL Containers + RHEL Host = Guaranteed Portability Across Any Infrastructure



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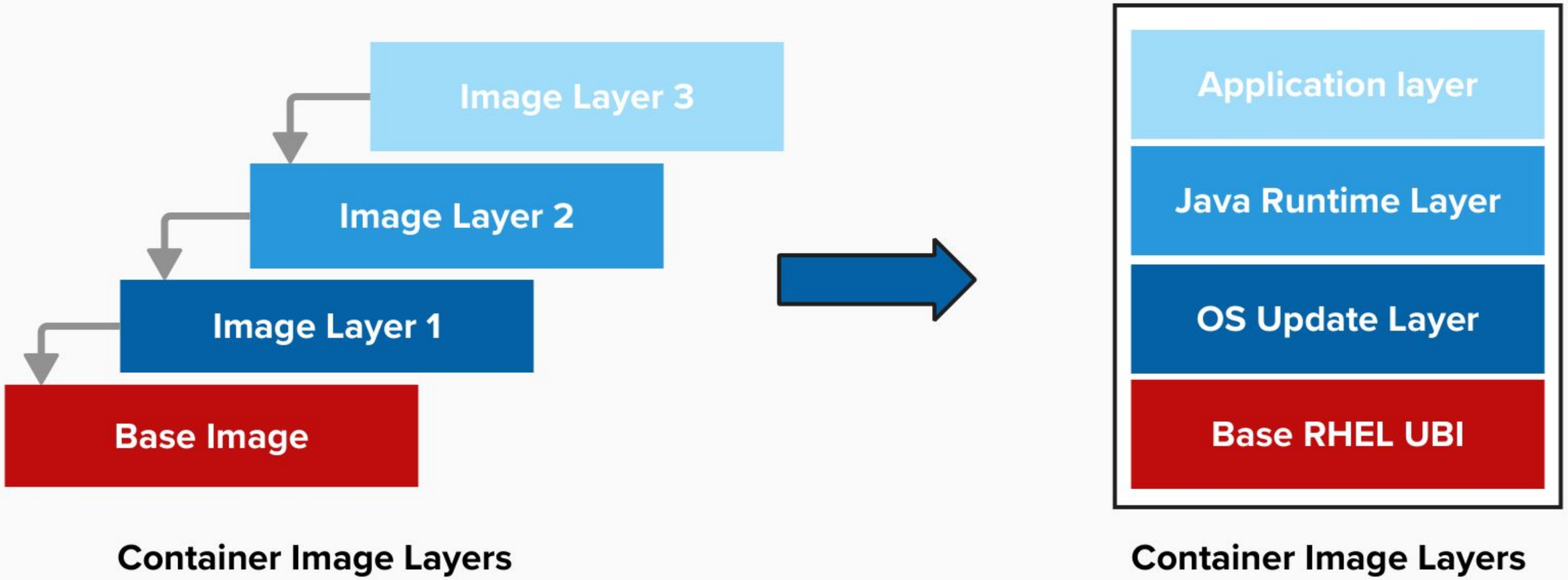
2. Application Portability with Containers

RHEL Containers + RHEL Host = Guaranteed Portability Across Any Infrastructure



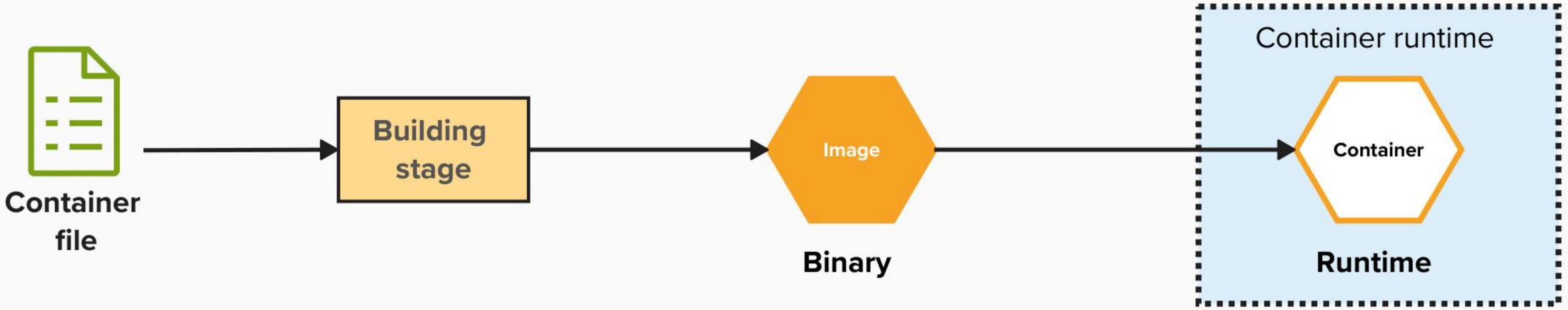
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2. Rapid Security Patching Using Container Image Layering



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2. Containers are created from container images



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containers are created from container images

Enough Theory!

Let’s build some containers 🛠️

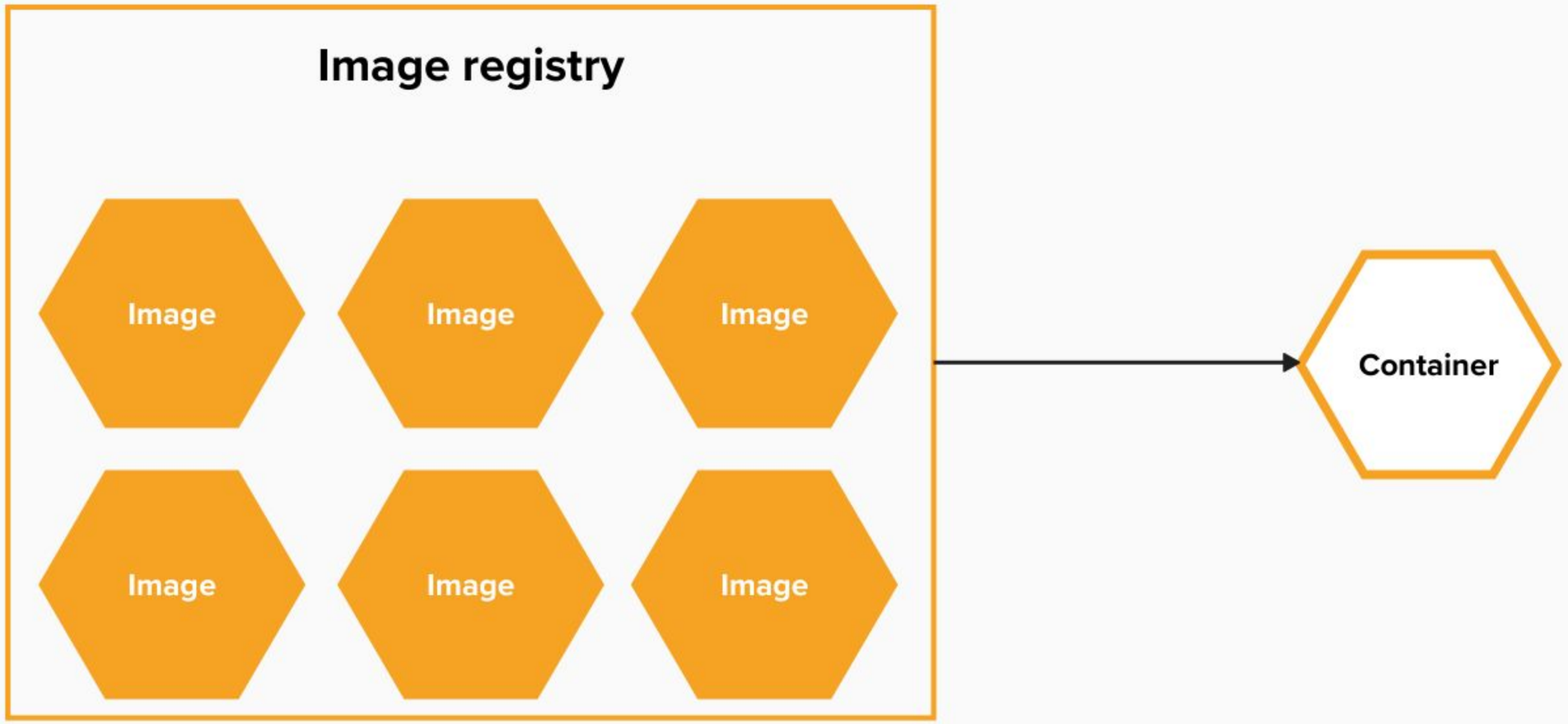
containers are created from container images

CONTAINER IMAGE REGISTRY



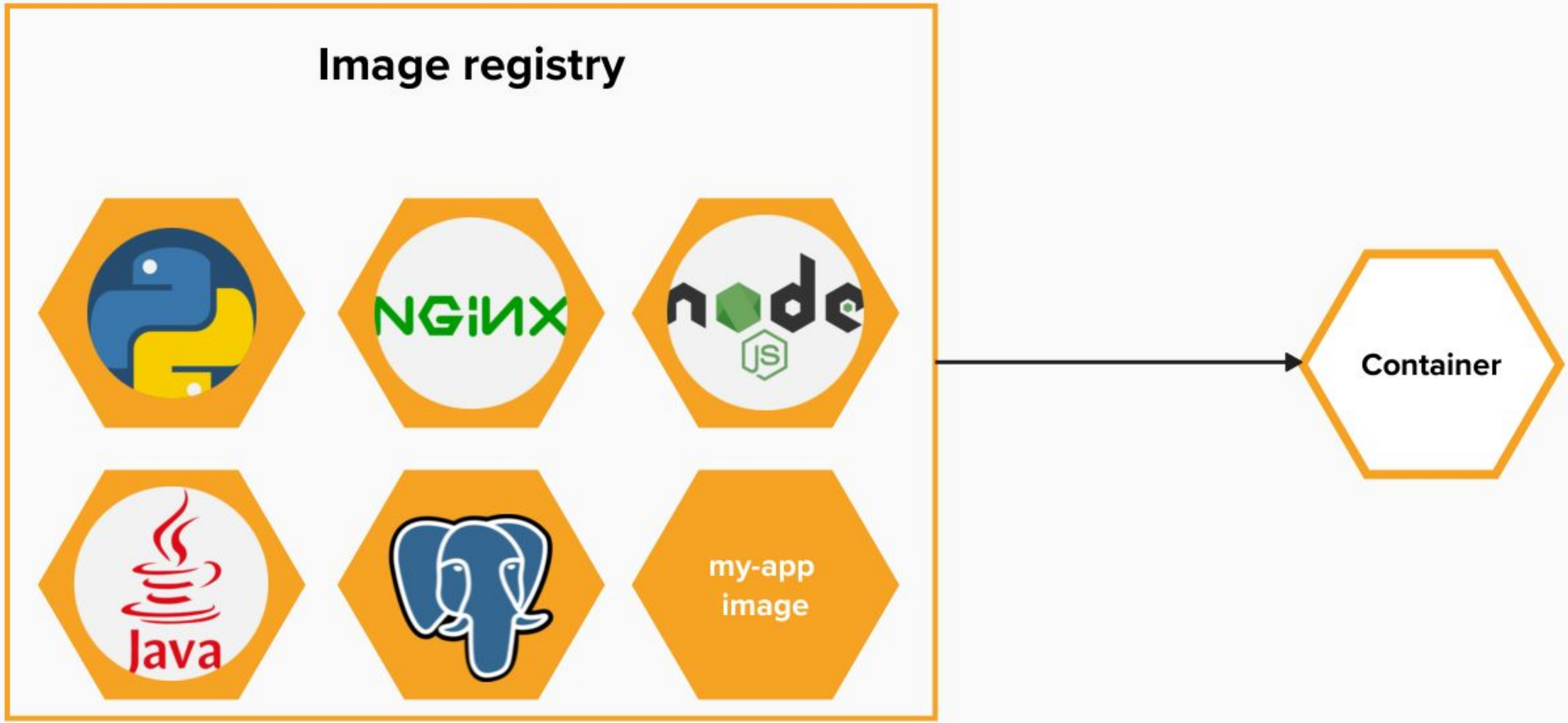


Container images are stored in an image registry



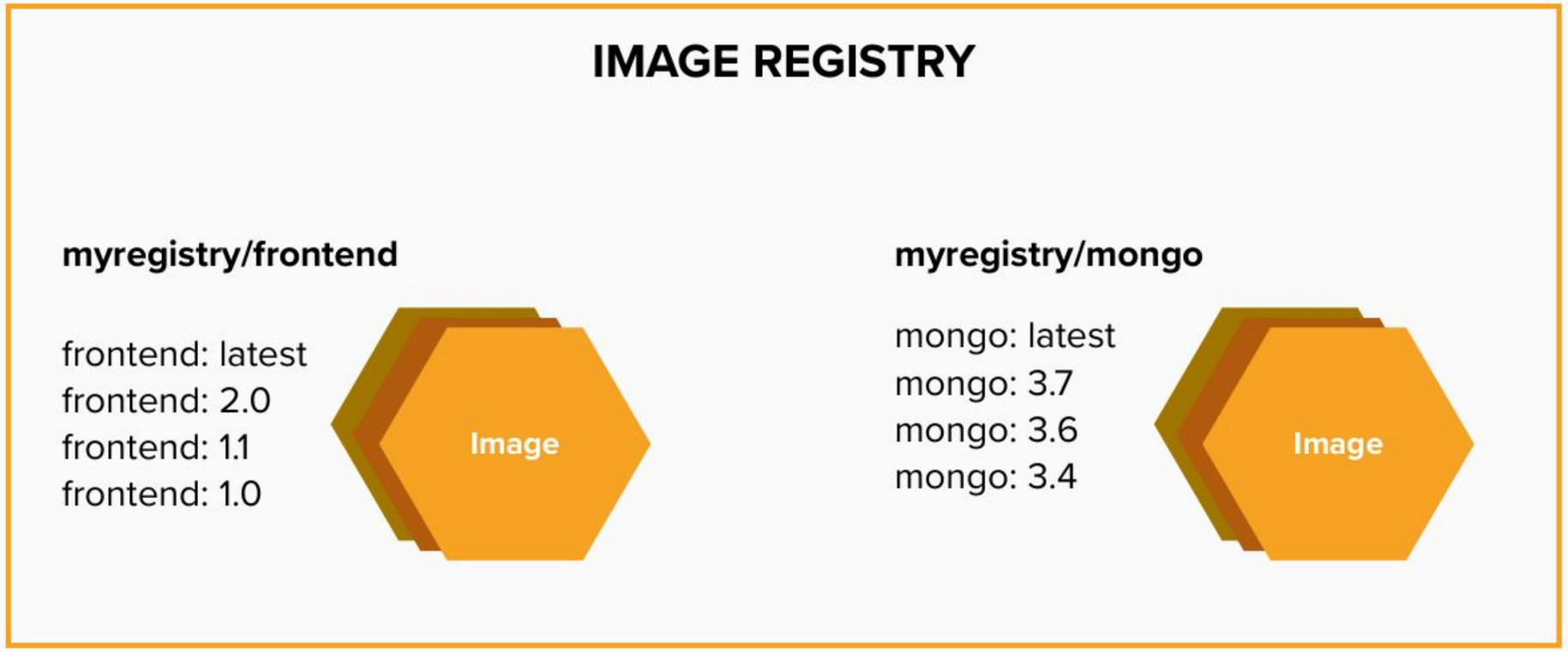
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Container images are stored in an image registry



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An image repository contains all versions of an image in the image registry



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containers are created from container images

You built your image, now let’s publish it.

Container Registry Hands-on Session

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containers are created from container images

Questions



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